

Replication: Leveraging Large-Scale Granular Single-Source Data for TV Advertising

An Identification Strategy with a Control Function (Hsieh, Du, and Lu, Marketing Science
2026)

Weiheng Zhang

September 07, 2026

Contents

1	The paper in one page	2
2	The design, step by step	2
2.1	The DGP the identification rests on	2
2.2	The instrument	3
2.3	The response model and the control function	3
3	What the replication package allows	3
4	Benchmark: the authors' scripts, rerun and matched	4
4.1	The scripts, verbatim	4
4.2	Step 4 with the package	5
5	A show level for the authors' DGP	6
5.1	What is added, and why	6
5.2	Table 1: descriptives	7
5.3	Figure 1: model-free evidence	7
6	Validating the instrument	8
6.1	Figure 2: is the focal brand scheduled like everyone else?	8
6.2	Figure 3: expected exposure as an area under the curve	8
6.3	Table 2 and the falsification checks	9
7	First stage	10
8	Second stage	10
8.1	Table 4: six columns, with and without the control function	10
8.2	Elasticities: the headline number	13
9	Ad responsiveness and purchase history	13
9.1	Table 5: observed household heterogeneity	14
10	The robustness checks the paper runs	15
10.1	The two decay rates (footnote 17)	15
10.2	Collinearity of the two stocks (footnote 18)	16
10.3	Placebo: another brand's ads (footnote 19, Online Appendix E.4)	16
10.4	Functional form of the history terms (Online Appendix E.1)	16

11 Extensions	17
11.1 The instrument is a recentered instrument	17
11.2 What the diagnostics catch when the assumptions fail	17
11.3 Estimators in probability units	18
11.4 Inference for the two-step estimator	19
12 What the replication teaches about the workflow	20
13 References	20

1 The paper in one page

The question is whether linear TV advertising causes purchases, and how the effect changes with a household’s purchase history. Hsieh, Du, and Lu (2026) link second-by-second TV viewing from 1.4 million smart-TV households to the daily orders of a food delivery platform over 133 days. TV advertising cannot be randomized at the household level, so the answer must come from observational exposure data.

Observed exposure is endogenous for two reasons, and controls cannot fix either. Targeting bias: the brand buys shows whose audiences already order more. Activity bias: households that watch more TV, or watch on a given day, differ in demand; the same unexpected long workday lowers TV time and raises the chance of ordering delivery. Both operate through household-day unobservables, which no set of controls absorbs.

The instrument comes from an institutional feature: networks assign within-show ad slots quasi-randomly across advertisers. A household that watches only part of a targeted show sees the focal ad only if the ad happens to air during the watched segment. Expected exposure is the integral of the network’s ad-timing density over the watched segment, $P_{is} = \int_{V_{iew_{is}}} l_s(x) dx$. The instrument is realized minus expected exposure, $IV_{is} = A_{is} - A_s^b P_{is}$, summed over the day’s shows. It is nonzero only for partial viewers of targeted shows, mean zero, positively correlated with exposure, and uncorrelated with the confounder (Proposition 1) and with every endogenous input, the targeting decision, the viewing probability, and their product (Proposition 2). The last property gives falsification checks from observables.

The purchase model is a probit with an ad stock and a control-function stock. Utility is $U_{it} = \alpha_{it} + \beta_{it} AS_{it} + \gamma X_{it} + \delta CFS_{it} + \varepsilon_{it}$ with $AS_{it} = \sum_l \lambda_A^l A_{i,t-l}$. Baseline propensity α_{it} and ad responsiveness β_{it} are quadratic in log purchase frequency and log recency after conversion, shift with household traits, and carry random effects. The first stage regresses daily exposure on the instrument and the controls; its residual is accumulated into $CFS_{it} = \sum_l \lambda_{CF}^l \hat{\varepsilon}_{i,t-l}$. The coefficient δ tests exogeneity. Standard errors come from 50 household bootstraps, and the two decay rates from an out-of-sample grid search.

Two findings. Ignoring endogeneity overstates ad elasticities by 55 percent (same-day 0.072 against 0.045, 30-day 0.222 against 0.143). Ad responsiveness is an inverted U in purchase frequency and a U in recency, so early repeat buyers and lapsed buyers respond most.

The replication package holds no data. The authors can share only a simulated data-generating process (DGP) with a known truth and the estimation code. This note reruns their scripts, matches them with the package’s control-function functions, adds the show level that their scripts omit so that every table and figure of the paper has a counterpart in the published layout, and then asks what the design would look like under violations of its assumptions.

2 The design, step by step

2.1 The DGP the identification rests on

Purchases respond to exposure and to a demand shock the analyst does not see. For household i on day t , $Y_{it} = f(\alpha + \beta A_{it} + \gamma X_{it} + u_{it})$ with $A_{it} = \sum_{s \in S_t} A_{is}$ (Equation 1). Endogeneity means A_{it} is correlated with u_{it} .

Exposure is the product of three decisions made by three parties. The brand targets show s , $A_s^b = f_b(X_s, u_s)$ (Equation 2); the household watches segments $View_{is} = f_v(X_{is}, u_{is}) \subseteq [0, 1]$ (Equation 3); the network draws the airing time $T_s^b \sim l_s(x)$ (Equation 4). Exposure is $A_{is} = A_s^b 1(T_s^b \in View_{is})$ (Equation 5). The first two decisions may depend on the demand shock, which is where targeting bias and activity bias enter. The third does not, which is the exogenous variation.

Two assumptions make the third decision usable. Quasi-random ordering: l_s does not depend on the brand or the household, so it can be estimated from every brand’s ads on that network. Nonstrategic viewership: $View_{is}$ does not depend on T_s^b , so a household does not switch channels because the focal ad is about to air. Each has an empirical check: compare focal and nonfocal airing-time distributions, and compare ad-completion rates for focal and nonfocal ads.

2.2 The instrument

Decompose realized exposure into an expected part and a shifter. $A_{is} = A_s^b P_{is} + A_s^b (1(T_s^b \in View_{is}) - P_{is})$ (Equation 7). The first term collects everything endogenous; the second is a centered Bernoulli draw, $1 - P_{is}$ with probability P_{is} and $-P_{is}$ otherwise, whose mean is zero given the targeting and viewing decisions. That second term is the instrument (Equation 8).

Why the mean-zero property matters for practice. Because $E[IV_{is} | A_s^b, View_{is}] = 0$, the instrument is uncorrelated with u_{is} , with A_s^b , with P_{is} , and with $A_s^b P_{is}$. The first is exogeneity and cannot be tested; the other three can, and a rejection of any of them means expected exposure was mis-estimated. The instrument is a residual from a known assignment process, not from a fitted model of exposure, so its validity does not depend on a propensity model being right.

2.3 The response model and the control function

Ad effects accumulate and depend on purchase history. Equations 9 to 15: an ad stock with daily decay λ_A ; α_{it} and β_{it} each equal to a base level, a shift after the first purchase, a quadratic in log frequency and log recency, a linear index in household traits Z_i , and a household random effect. The log and squared-log terms let both curves bend, which is what Figure 4 of the paper reports.

A nonlinear second stage needs a control function, not two-stage least squares. Equations 16 to 19: regress A_{it} on IV_{it} and the controls, keep the residual $\hat{\varepsilon}_{it}^f$, accumulate it with decay λ_{CF} into CFS_{it} , and add CFS_{it} to the probit. Conditioning on the residual removes the part of the demand shock that is correlated with the ad stock; $\delta = 0$ is the exogeneity test (Wooldridge 2015). The stock is needed because the endogenous regressor is itself a stock: if the confounder persists, so must its control, and the two decays need not be equal.

Inference must account for the generated regressor. The second stage treats CFS_{it} as data. The paper resamples households, 50 times, and re-runs both stages.

3 What the replication package allows

The data are under a nondisclosure agreement; the package simulates them. Table 1 lists the four scripts. Their DGP is a household-day panel with a known truth. It keeps the two biases of the paper, an unobservable that drives targeting, viewing, and purchase, but it has no show level: no airing-time density, no viewing segments, no other brands’ ads.

Three points where the scripts and the paper part ways. First, the scripts’ expected exposure uses the targeting probabilities, which are functions of the unobservable; an analyst never sees them. The paper’s expected treatment conditions on the realized targeting decision and the watched portion. Both are mean zero given the unobservable, so both are valid in the simulation, but only the second can be computed from data. Second, the bootstrap keeps each sampled household once (`filter(hh %in% boot_hh)`), so every “bootstrap” sample is a 63 percent subsample without replacement, and its standard deviation is about

Table 1: The replication package and its role in this note.

Script	Content	Used for
Step1 - AdExposureSimulation.R	25,000 households x 150 days; u_{endo} drives the targeting probability of three shows a day and the watched portion; exposure = Bernoulli(portion) x targeting; expected exposure; ad stock with decay 0.7	benchmark panel, rerun with seed 1107
Step2 - PurchaseSimulation.R	probit utility with $\alpha_0 = -2.5$, $\beta_0 = 0.012$, state terms, a 30-day promotion after the first purchase, competitor spend, and a confounder that accumulates u_{endo} with decay 0.95	benchmark outcomes, rerun with seed 123
Step3 - BootstrapEstimate.R	50 household resamples; first stage, control-function stock with decay 0.95, probit	benchmark standard errors
Step4 - ModelEstimation.R	naive probit and control-function probit; a table with the true parameters	benchmark point estimates

$\sqrt{1/0.632 - 1} = 0.76$ of a cluster bootstrap in expectation. Third, the scripts have no show level, so Figures 2 and 3 and the household-show half of Table 2 cannot be produced from them.

What this note does with each. Section 4 reruns the scripts and matches them with `cf_residuals()` and a cluster bootstrap that keeps duplicates, reporting both instruments and both standard errors. Sections 5 to 9 add a show level, networks with their own airing-time densities, viewing segments, nonfocal ads, on top of the authors' purchase model, and reproduce the paper's tables and figures in their layouts. Section 10 reruns the paper's robustness checks, and Section 11 asks what the diagnostics catch when the assumptions fail.

4 Benchmark: the authors' scripts, rerun and matched

4.1 The scripts, verbatim

The helper file reproduces Steps 1 and 2 line by line, with seeds 1107 and 123. Random streams inside `future_lapply(future.seed = TRUE)` do not depend on the number of workers, so the panel equals the one the authors would obtain. The vectorized purchase simulation used later in this note is checked against the authors' household loop on the same panel.

```
fs_file <- sim_file("fs_ready"); ss_file <- sim_file("ss_ready")
if (file.exists(ss_file)) {
  ss <- readRDS(ss_file)
} else {
  fs <- authors_step1(N_hh = n_hh, T_day = n_days, workers = n_threads)
  saveRDS(fs, fs_file)
  ss <- authors_step2(fs, workers = n_threads)
  saveRDS(ss, ss_file)
}
ours <- simulate_purchases(ss[order(ss$hh, ss$t), c("hh", "t", "A", "A_stock", "u_endo", "u_exo", "comp")], purchase_par(FALSE,
c(rows = nrow(ss), purchase_rate = mean(ss$purchase), converted = mean(tapply(ss$purchase, ss$hh, max)),
  same_purchases = all(ours$purchase == ss$purchase),
  same_states = all(ours$freq == ss$freq & ours$rec == ss$rec & ours$promo1m == ss$promo1m))
#>      rows purchase_rate converted same_purchases same_states
#> 3.750000e+06 1.437085e-01 6.350400e-01 1.000000e+00 1.000000e+00
```

The authors' purchase parameters convert 63.5 percent of households within 150 days, against 3.8 percent in the paper; the daily conversion rate of an unconverted household, $\Phi(-2.5) = 0.62$ percent, is fifty times the paper's 0.012 percent. The confounder enters utility with weight 0.005 on a stock whose standard deviation is about 3.2, so the endogeneity is mild. We keep every value as shipped.

4.2 Step 4 with the package

The first stage is `cf_residuals()`, the stock is a two-line recursion, and the second stage is `glm()`. `cf_residuals()` returns the data with the residual appended and the first-stage strength of `iv_first_stage()` as an attribute. The stock CFS_{it} is accumulated within household with `adstock_by()`, defined in the helper file.

```
adstock <- function(x, lambda) as.numeric(stats::filter(x, lambda, method = "recursive"))
adstock_by <- function(x, id, lambda) as.numeric(stats::ave(x, id, FUN = function(v) adstock(v, lambda)))

s4 <- authors_step4(ss)
bench <- add_states(ss)
spec_bench <- col_spec(lambda_A = 0.7, cf = TRUE, lambda_CF = 0.95, interactions = TRUE, traits = FALSE)
ours_bench <- fit_column(bench, spec_bench)
fs_strength <- attr(ours_bench$data, "first_stage")$strength
cf_authors <- rename_authors(coef(s4$full)); cf_naive <- rename_authors(coef(s4$naive))
terms_b <- names(cf_authors)
ours_cf <- coef(ours_bench$fit)[terms_b]
bench_model_se <- summary(ours_bench$fit)$coefficients[terms_b, "Std. Error"]
c(max_abs_diff = max(abs(cf_authors - ours_cf)), F_first_stage = fs_strength$F, F_robust = fs_strength$F_robust)
#> max_abs_diff F_first_stage F_robust
#> 9.436896e-16 2.186849e+06 1.197535e+06
rm(s4, ours_bench); invisible(gc())

boot_file <- sim_file(paste0("bench_boot_", n_boot))
if (file.exists(boot_file)) {
  bb <- readRDS(boot_file)
} else {
  b_authors5 <- authors_step3(ss, B = min(5L, n_boot), seed = 1107, workers = n_threads_boot)
  bench_cols <- c("hh", "t", "purchase", "A", "IV", controls_base, "freq", "rec")
  b_unique <- cluster_bootstrap(bench[, bench_cols], spec_bench, n_boot = n_boot, seed = 1107, unique_only = TRUE,
    workers = n_threads_boot)
  b_cluster <- cluster_bootstrap(bench[, bench_cols], spec_bench, n_boot = n_boot, seed = 1107, unique_only = FALSE,
    workers = n_threads_boot)
  bb <- list(authors5 = b_authors5, unique = b_unique, cluster = b_cluster)
  saveRDS(bb, boot_file)
}
auth_draws <- rename_authors(bb$authors5$draws)
n5 <- min(nrow(auth_draws), nrow(bb$unique$draws))
c(first_draws_max_diff = max(abs(auth_draws[seq_len(n5), ] - bb$unique$draws[seq_len(n5), colnames(auth_draws)])))
#> first_draws_max_diff
#> 3.530509e-13
```

Table 2: The authors’ Step 4 table on their DGP (25,000 households, 150 days): true parameters, naive probit, control-function probit, and three standard errors. The authors’ bootstrap keeps each sampled household once; the cluster bootstrap keeps duplicates. 50 draws each.

Term	True	Naive	Control function	Model SE	Authors’ bootstrap SE	Cluster bootstrap SE	t (cluster)
Intercept	-2.500	-2.5156	-2.5075	0.0057	0.0049	0.0062	-404.58
Ad stock	0.012	0.0186	0.0135	0.0026	0.0024	0.0029	4.57
Control function	—	—	0.0056	0.0006	0.0004	0.0006	9.42
Post conversion	1.500	1.5396	1.5402	0.0168	0.0127	0.0180	85.40
Frequency (log)	0.450	0.4180	0.4180	0.0123	0.0095	0.0122	34.35
Frequency (log) sq.	-0.060	-0.0535	-0.0535	0.0024	0.0018	0.0023	-23.09
Recency (log)	-0.100	-0.1069	-0.1069	0.0073	0.0057	0.0070	-15.17
Recency (log) sq.	-0.030	-0.0288	-0.0287	0.0036	0.0027	0.0036	-8.05
First month post-conversion promotion	0.200	0.2017	0.2014	0.0040	0.0030	0.0045	45.00
Competitor ad spend	0.020	0.0234	0.0235	0.0049	0.0038	0.0051	4.63
Ad stock x post conversion	0.000	-0.0210	-0.0211	0.0079	0.0057	0.0085	-2.47
Ad stock x freq (log)	0.020	0.0369	0.0368	0.0062	0.0050	0.0067	5.50
Ad stock x freq (log) sq.	-0.010	-0.0133	-0.0133	0.0012	0.0010	0.0014	-9.60
Ad stock x rec (log)	0.000	0.0030	0.0030	0.0037	0.0032	0.0038	0.77
Ad stock x rec (log) sq.	0.005	0.0051	0.0051	0.0018	0.0015	0.0019	2.68

What the benchmark shows. The package reproduces the authors’ point estimates to 9.4e-16, and its `unique_only` bootstrap reproduces their first five draws exactly, so the 50-draw column labelled “authors’ bootstrap” is what their Step 3 returns. The naive ad-stock coefficient is 0.0186 against a truth of 0.012

and a corrected 0.0135. The control-function coefficient is 0.0056 with cluster t statistic 9.42. The authors' bootstrap standard errors average 0.76 of the cluster bootstrap's, close to the 0.76 that a 63 percent subsample implies, and the model-based standard errors that treat the stock as data are smaller still for the ad-stock terms. Table 3 compares the scripts' oracle instrument with the paper's.

Table 3: Two instruments on the authors' DGP. The scripts subtract the expected exposure computed from the targeting probabilities; the paper subtracts the expected exposure given the realized targeting decisions.

Instrument	Mean	Corr with A	Corr with u_{endo}	First-stage F	β_0	δ
Scripts: $A - \sum_k P_k p_k^{target}$ (oracle)	0.0003	0.607	-0.0001	2,186,849	0.0135	0.0056
Paper: $A - \sum_k A_k^b P_k$	0.0002	0.314	0.0002	409,042	0.0135	0.0039

Both instruments are mean zero and orthogonal to the unobservable, and both correct the naive estimate. The paper's version is the one an analyst can build, and the rest of the note uses it.

5 A show level for the authors' DGP

5.1 What is added, and why

The paper's construction needs shows, densities, and segments. The helper simulates, for each household-day, three shows on three networks. Each network has its own airing-time density on $[0, 1]$: trimodal like MTV in the paper's Figure 2, bimodal, and close to uniform. The brand targets each show with the authors' probabilities, rising in the household-day shock. The household watches a segment whose length follows the authors' rule (a 30 percent chance of not watching, otherwise a length rising in the shock) plus two standardized traits, average viewing and sports viewing, the latter acting on the second network only. The focal ad airs at a time drawn from the network's density; nonfocal ads are drawn from the same density. Exposure follows Equation 5. Expected exposure integrates the empirical distribution of nonfocal ads over the segment, and the instrument is Equation 8. A second brand with no effect on purchases is exposed through the same viewing process for the placebo check. The purchase model is the authors' with two additions for Table 4 columns 5 and 6 and Table 5: the two traits enter α_{it} ($-0.05, 0.03$) and β_{it} ($-0.004, 0.004$), and household random effects have $\sigma_1 = 0.2, \sigma_2 = 0.01, \rho = -0.3$.

Two of the authors' values are changed, and why. The intercept is -3.0 instead of -2.5 : with -2.5 two thirds of households convert within 150 days and repeat buyers reach purchase frequencies at which the quadratic in log frequency turns the ad effect negative on average; with -3.0 about a quarter convert, still far above the paper's 3.8 percent but with an average ad effect of the paper's sign. The weight of the confounder is 0.015 instead of 0.005: with the authors' value the naive bias in β_0 is about 0.003, below the standard error at this sample size, so the comparison that Table 4 is built to show would sit inside sampling noise. Both changes are to the purchase model only; the exposure process is the authors' with a show level added, and the benchmark of Section 4 uses the authors' values throughout.

```
show_file <- sim_file("show_panel")
if (file.exists(show_file)) {
  sim <- readRDS(show_file)
} else {
  par_b <- purchase_par(traits = TRUE, random = TRUE, a0 = -3.0, delta = 0.015)
  sim <- simulate_exposure(n_hh = n_hh, n_days = n_days, seed = 1107, par = par_b)
  sim$panel <- add_states(simulate_purchases(sim$panel, sim$par))
  saveRDS(sim, show_file)
}
pan <- sim$panel
set.seed(1107)
sub_hh <- sort(sample(unique(pan$hh), min(5000L, n_hh)))
pan_sub <- pan[pan$hh %in% sub_hh, ]
c(rows = nrow(pan), purchase_rate = mean(pan$purchase), converted = mean(tapply(pan$purchase, pan$hh, max)),
  exposures_per_day = mean(pan$A), partial_share = sum(pan$n_partial) / sum(pan$n_target))
#>           rows      purchase_rate      converted exposures_per_day
```

```
#>      3.750000e+06      2.718720e-02      2.281600e-01      5.041416e-01
#>      partial_share
#>      2.664362e-01
```

5.2 Table 1: descriptives

The paper’s Table 1 tells the reader how rare exposures and purchases are. Ours reports the same five quantities for the simulated panel; show length is one hour.

Table 4: Descriptive statistics (the layout of Table 1). Simulated TV viewing and purchase behavior for 25,000 households over 150 days.

	Mean	SD	5%	25%	50%	75%	95%
TV viewing per household per day (hr)	0.86	0.99	0.00	0.00	0.42	1.71	3.00
Targeted show viewing per household per day (hr)	0.49	0.74	0.00	0.00	0.00	0.97	2.00
Number of focal brand ad exposures per household per day	0.50	0.79	0.00	0.00	0.00	1.00	2.00
Purchase frequency per converted household	17.87	13.10	1.00	7.00	15.00	27.00	42.00
Interpurchase time per converted household (days)	4.25	5.59	1.00	1.00	2.00	5.00	13.00

Note. 5,704 households (22.8 percent) converted.

5.3 Figure 1: model-free evidence

The paper plots purchase probability against frequency, recency, and same-day exposures before fitting anything. The first two panels motivate the state-dependence terms; the third is the naive association that the instrument is meant to shrink. The same three panels for the simulated panel:

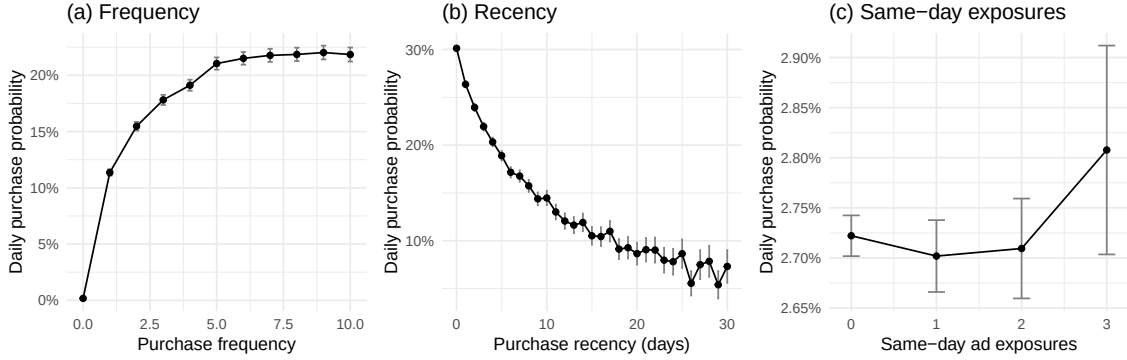


Figure 1: Daily purchase probability by prior purchase frequency, purchase recency, and same-day ad exposures (the layout of Figure 1). Means with 95 percent confidence intervals.

Converted households buy every 4.3 days here against 11.3 in the paper, and 17.9 times over the window against 2.4, because the post-conversion shift of 1.5 in the authors’ utility is large relative to the paper’s baseline. Exposures per day (0.50) are the authors’ and about four times the paper’s 0.14.

Panel (a) rises with frequency because of habit formation in the DGP ($\alpha_{2,1} = 0.45$, $\alpha_{2,2} = -0.06$). Panel (b) falls with recency ($\alpha_{2,3} = -0.10$, $\alpha_{2,4} = -0.03$); the paper’s weekly spikes are absent because the DGP has no day-of-week reordering. Panel (c) is positive, and part of it is activity bias.

6 Validating the instrument

6.1 Figure 2: is the focal brand scheduled like everyone else?

The quasi-random ordering assumption is checked by comparing airing-time distributions. If the network gave the focal brand a special slot, the focal distribution would differ from the nonfocal one, and the density estimated from all ads would mis-state expected exposure. The paper shows MTV; we show the first network, whose density is trimodal by construction, and the Kolmogorov-Smirnov test for every network (Online Appendix E.2).

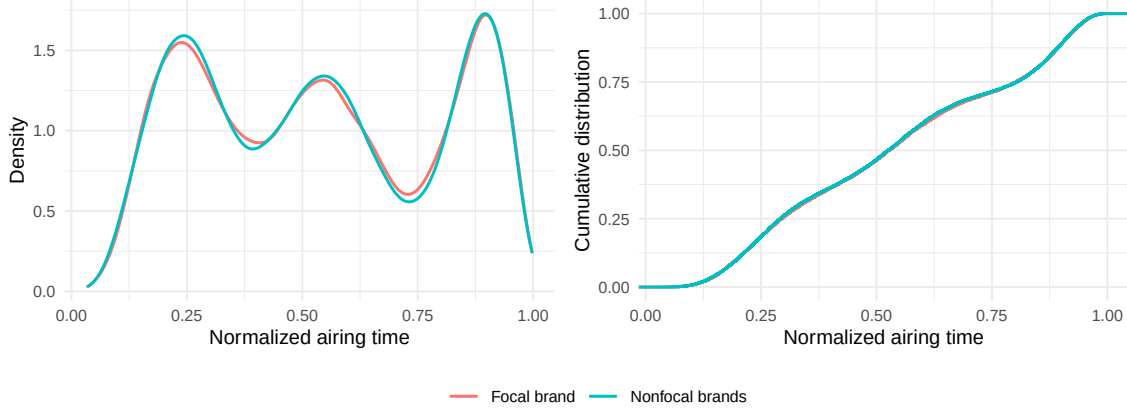


Figure 2: Within-show airing-time distributions of focal and nonfocal ads on the first network (the layout of Figure 2). Left: densities. Right: cumulative distributions. Airing times are normalized by show duration.

Table 5: Kolmogorov-Smirnov tests of focal against nonfocal airing-time distributions by network (the check of Online Appendix E.2).

Network	Focal ads	Nonfocal ads	KS statistic	p-value
Network 1	22500	20000	0.010	0.259
Network 2	22500	20000	0.009	0.365
Network 3	22500	20000	0.011	0.170

The focal and nonfocal distributions coincide on every network because the DGP draws both from the same density; the KS p-values are 0.26, 0.37, 0.17. The paper reports 0.545 for MTV. Section 11.2 shows what the test does when the assumption fails.

6.2 Figure 3: expected exposure as an area under the curve

The paper’s Figure 3 turns the definition of P_{is} into a picture. The left panel is the network’s density; the right panel shades the household’s viewing window and reports the area, which is the probability that a focal ad on that network airs while the household is watching.

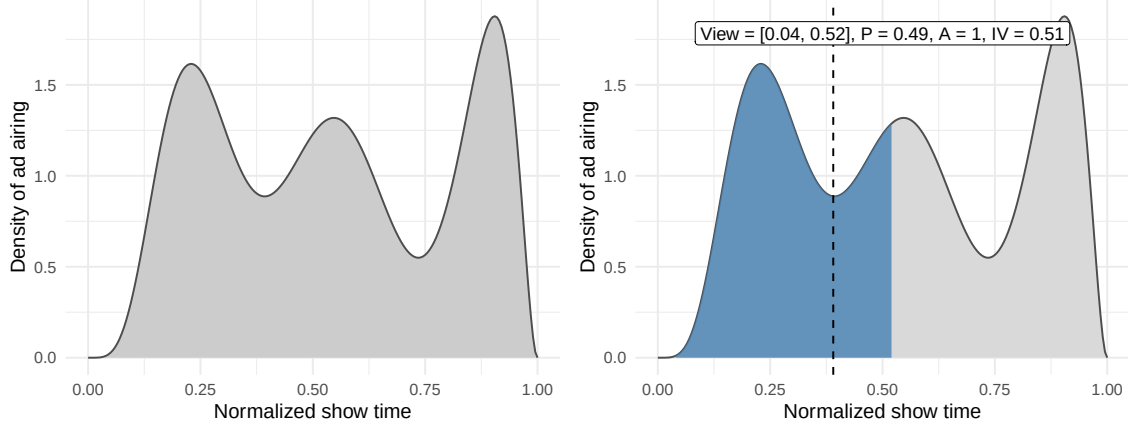


Figure 3: Airing-time density of the first network and the expected exposure of a household that watched part of a targeted show (the layout of Figure 3). The shaded area is the estimated probability of exposure, the dashed line the realized airing time of the focal ad.

The household in the right panel watched 49 percent of a targeted show and had an expected exposure of 0.49; its realized exposure was 1, so its instrument is 0.51. The uniform-density shortcut would have used 0.49, the viewed share, instead.

6.3 Table 2 and the falsification checks

Table 2 of the paper reports the moments that Propositions 1 and 2 predict. Realized and expected exposure should have the same mean, so the instrument is mean zero; the instrument should correlate with realized exposure and not with the targeting decision, the exposure probability, or their product. Household-show rows are those in which a household watched any part of a targeted show, as in the paper; our household-show quantiles use a 2 percent sample of those rows while the means, standard deviations, and correlations use all of them.

Table 6: Summary statistics of realized ad exposure, expected ad exposure, and the instrument (the layout of Table 2).

	Mean	SD	5%	25%	50%	75%	95%
Household-show level							
Realized ad exposure (A_{is})	0.767	0.423	0.000	1.000	1.000	1.000	1.000
Expected ad exposure ($A_s^b \hat{P}_{is}$)	0.767	0.313	0.111	0.536	0.997	1.000	1.000
Proposed IV ($\widehat{IV}_{is}$)	-0.000	0.284	-0.551	0.000	0.000	0.012	0.536
Household-day level							
Realized ad exposure (A_{it})	0.504	0.789	0.000	0.000	0.000	1.000	2.000
Expected ad exposure ($\sum_s A_s^b \hat{P}_{is}$)	0.504	0.754	0.000	0.000	0.000	1.000	2.000
Proposed IV ($\widehat{IV}_{it}$)	-0.000	0.230	-0.392	0.000	0.000	0.000	0.388

Note. Based on 2,465,867 household-show observations in which the household watched part of a targeted show, and 3,750,000 household-day observations.

Reading the checks. Realized and expected exposure have the same mean to three decimals, so the instrument is mean zero; it correlates with realized exposure at 0.29 on the day level (0.36 in the paper) and with nothing endogenous. The last row cannot be computed with real data and is the reason the design works: the instrument does not move with the demand shock even though exposure does ($\text{corr}(A_{it}, u_{endo}) = 0.62$).

Table 7: Relevance and falsification checks of the instrument (Propositions 1 and 2), with the paper’s values.

Check	Household-show	Household-day	Paper
Mean of the IV (t statistic)	-0.0003 (-1.51)	-0.0002 (-1.51)	-0.002; -0.004
corr(IV, realized exposure)	0.671	0.292	0.42; 0.36
corr(IV, targeting decision A_s^b)	-0.0005	-0.0003	-0.001; -0.006
corr(IV, exposure probability $\hat{P}_{is}$)	0.0004	—	-0.001; -0.007
corr(IV, expected exposure $A_s^b \hat{P}_{is}$)	0.0004	-0.0004	-0.009; -0.009
corr(IV, u_{endo}), unobservable in practice	0.0002	-0.0002	not observable

7 First stage

Table 3 shows that the instrument moves exposure one for one and that purchase history does not, once the instrument is in the regression. The coefficient on the instrument is close to one because $A_{it} = \text{expected} + IV_{it}$ and the expected part is absorbed by the controls only in part. `cf_residuals()` runs this regression and returns the residual and the strength statistics.

Table 8: Parameter estimates from the first-stage model (the layout of Table 3). Standard errors clustered by household.

Model component	Parameter	Estimate	Standard error
Intercept	α_0^f	0.5051***	0.0010
$\widehat{IV}_{it}$	ϕ^f	0.9989***	0.0015
Post conversion	$\alpha_{2,0}^f$	0.0082	0.0089
Frequency (log)	$\alpha_{2,1}^f$	-0.0039	0.0069
Frequency (log) squared	$\alpha_{2,2}^f$	0.0006	0.0015
Recency (log)	$\alpha_{2,3}^f$	-0.0006	0.0029
Recency (log) squared	$\alpha_{2,4}^f$	-0.0006	0.0009
First month post-conversion promotion	γ_1^f	-0.0042	0.0037
Competitor ad spend	γ_2^f	-0.0013	0.0019
Average TV viewing	α_1^f	0.0858***	0.0004
Sports viewing	α_1^f	0.0283***	0.0004

Note. The DV is daily household focal brand ad exposures. First-stage F = 353,612 (robust F = 431,337). ***p < 0.01; **p < 0.05; *p < 0.1.

The first-stage F is 353,612 (the paper: 1,902,996), so weak-instrument inference is not a concern here; on a 5,000-household subsample the Anderson-Rubin 95 percent set for the linear-probability effect of exposure, [-0.0023, 0.0005], equals the Wald interval [-0.0023, 0.0005]. The purchase-history terms do not predict exposure given the instrument, as in the paper’s Table 3, because the DGP routes all endogeneity through the unobservable rather than through the observed states.

8 Second stage

8.1 Table 4: six columns, with and without the control function

The paper builds the model up in three steps and adds the control function at each step. Columns 1 and 2 have no carryover; columns 3 and 4 add the two decays; columns 5 and 6 add the moderation of ad responsiveness by purchase history and by household traits and the household random effects. Comparing each pair isolates what the control function does at that level of richness. Columns 5 and 6 are estimated with a household random intercept and random slope on the ad stock through `lme4::glmer()` (Laplace approximation, `nAGQ = 0`) instead of the paper’s simulated maximum likelihood.

Bootstrap standard errors use 50 household draws with duplicates kept; for column 6 the coefficient rows are bootstrapped without the random effects (each random-effects fit takes about twenty minutes at this size) and the three random-effect rows on the 5,000-household subsample with 20 draws.

```
specs <- list(
  c1 = col_spec(0, FALSE, 0, FALSE, TRUE, FALSE),
  c2 = col_spec(0, TRUE, 0, FALSE, TRUE, FALSE),
  c3 = col_spec(0.7, FALSE, 0, FALSE, TRUE, FALSE),
  c4 = col_spec(0.7, TRUE, 0.95, FALSE, TRUE, FALSE),
  c5 = col_spec(0.7, FALSE, 0, TRUE, TRUE, TRUE, random = TRUE),
  c6 = col_spec(0.7, TRUE, 0.95, TRUE, TRUE, TRUE, random = TRUE)
)
fits_file <- sim_file("table4_fits")
if (file.exists(fits_file)) {
  t4 <- readRDS(fits_file)
} else {
  t4 <- lapply(specs, function(s) {
    f <- fit_column(pan, s)
    se <- if (inherits(f$fit, "merMod")) sqrt(diag(as.matrix(vcov(f$fit)))) else summary(f$fit)$coefficients[, 2]
    list(coef = coef_vector(f$fit), se = se)
  })
  saveRDS(t4, fits_file)
}
sapply(t4, function(x) x$coef[c("A_stock", "CF_stock")])
#>           c1           c2           c3           c4           c5           c6
#> A_stock 0.02055387 0.002054785 0.02457711 0.01063892 0.02737131 0.01142566
#> <NA>           NA 0.020254871           NA 0.01014235           NA 0.01209124

boot4_file <- sim_file(paste0("table4_boot_", n_boot))
if (file.exists(boot4_file)) {
  b4 <- readRDS(boot4_file)
} else {
  boot_cols <- c("hh", "t", "purchase", "A", "IV", controls_base, controls_traits, "freq", "rec")
  b4 <- list(
    c2 = cluster_bootstrap(pan[, boot_cols], specs$c2, n_boot = n_boot, seed = 1107, workers = n_threads_boot),
    c4 = cluster_bootstrap(pan[, boot_cols], specs$c4, n_boot = n_boot, seed = 1107, workers = n_threads_boot),
    c6 = cluster_bootstrap(pan[, boot_cols], modifyList(specs$c6, list(random = FALSE)), n_boot = n_boot, seed = 1107,
      workers = n_threads_boot),
    c6_re = cluster_bootstrap(pan_sub[, boot_cols], specs$c6, n_boot = min(n_boot, 20L), seed = 1107,
      workers = n_threads_boot)
  )
  b4$c6$se[c("sigma1", "sigma2", "rho")] <- b4$c6_re$se[c("sigma1", "sigma2", "rho")]
  saveRDS(b4, boot4_file)
}
sapply(b4[c("c2", "c4", "c6")], function(b) c(b$se[c("A_stock", "CF_stock")], failed = b$n_failed))
#>           c2           c4           c6
#> A_stock 0.009690427 0.002400668 0.003430316
#> CF_stock 0.010114672 0.001023834 0.001035876
#> failed 0.000000000 0.000000000 0.000000000
```

Table 9: Second-stage estimation results for the ad response model (the layout of Table 4), simulated panel of 25,000 households. Columns 1 and 2 assume no carryover, no evolution of ad responsiveness, and no unobserved heterogeneity; columns 3 and 4 add carryover; columns 5 and 6 add moderation by purchase history and household traits and household random effects. Columns 2, 4, and 6 use the control function; their standard errors come from 50 household bootstrap draws. ***p < 0.01; **p < 0.05; *p < 0.1.

		No carryover or moderation, without CF (1)		No carryover or moderation, with CF (2)		No moderation, without CF (3)		No moderation, with CF (4)		Full model, without CF (5)		Full model, with CF (6)		
		True	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Intercept	α_0	-2.500	-2.950***	0.006	-2.940***	0.008	-2.980***	0.007	-2.958***	0.007	-2.986***	0.009	-2.960***	0.009
Ad stock	β_0	0.012	0.021***	0.002	0.002	0.010	0.025***	0.002	0.011***	0.002	0.027***	0.004	0.011***	0.003
Control function	δ	—	—	—	0.020**	0.010	—	—	0.010***	0.001	—	—	0.012***	0.001
Post conversion	$\alpha_{2,0}$	1.500	1.459***	0.018	1.459***	0.018	1.457***	0.018	1.457***	0.018	1.444***	0.029	1.441***	0.026
Frequency (log)	$\alpha_{2,1}$	0.450	0.446***	0.014	0.446***	0.014	0.446***	0.014	0.446***	0.014	0.477***	0.024	0.477***	0.022
Frequency (log) sq.	$\alpha_{2,2}$	-0.060	-0.042***	0.003	-0.042***	0.003	-0.042***	0.003	-0.042***	0.003	-0.063***	0.005	-0.063***	0.005
Recency (log)	$\alpha_{2,3}$	-0.100	-0.124***	0.006	-0.124***	0.006	-0.124***	0.006	-0.124***	0.006	-0.113***	0.011	-0.113***	0.011
Recency (log) sq.	$\alpha_{2,4}$	-0.030	-0.025***	0.002	-0.025***	0.002	-0.025***	0.002	-0.024***	0.002	-0.028***	0.004	-0.027***	0.004
Ad stock x post conversion	$\beta_{2,0}$	0.000	—	—	—	—	—	—	—	—	0.003	0.013	0.005	0.011
Ad stock x freq (log)	$\beta_{2,1}$	0.020	—	—	—	—	—	—	—	—	0.014	0.011	0.013	0.010
Ad stock x freq (log) sq.	$\beta_{2,2}$	-0.010	—	—	—	—	—	—	—	—	-0.008***	0.003	-0.008***	0.002
Ad stock x rec (log)	$\beta_{2,3}$	0.000	—	—	—	—	—	—	—	—	0.003	0.005	0.003	0.006
Ad stock x rec (log) sq.	$\beta_{2,4}$	0.005	—	—	—	—	—	—	—	—	0.004**	0.002	0.004**	0.002
First month post-conversion promotion	γ_1	0.200	0.321***	0.007	0.321***	0.008	0.321***	0.007	0.320***	0.008	0.216***	0.007	0.215***	0.008
Competitor ad spend	γ_2	0.020	0.024***	0.009	0.024***	0.008	0.024***	0.009	0.024***	0.008	0.024***	0.009	0.024***	0.008
Carryover λ_A/λ_{CF}		0.7 / 0.95	0 / 0		0 / 0		0.7 / 0		0.7 / 0.95		0.7 / 0		0.7 / 0.95	
Household characteristics	α_1	Yes	Yes		Yes		Yes		Yes		Yes		Yes	
HH characteristics x ad stock	β_1	Yes	—		—		—		—		Yes		Yes	
Average TV viewing	α_1	-0.050	-0.042***	0.002	-0.041***	0.002	-0.048***	0.002	-0.044***	0.002	-0.051***	0.004	-0.046***	0.004
Sports viewing	α_1	0.030	0.029***	0.002	0.029***	0.002	0.027***	0.002	0.028***	0.002	0.032***	0.004	0.033***	0.003
Ad stock x average TV viewing	β_1	-0.004	—	—	—	—	—	—	—	—	-0.004**	0.002	-0.004**	0.002
Ad stock x sports viewing	β_1	0.004	—	—	—	—	—	—	—	—	0.002	0.002	0.002	0.001
Standard deviation (intercept)	σ_1	0.200	—	—	—	—	—	—	—	—	0.182	—	0.184	0.080
Standard deviation (ad stock)	σ_2	0.010	—	—	—	—	—	—	—	—	0.014	—	0.016	0.021
Rho	ρ	-0.300	—	—	—	—	—	—	—	—	-0.454	—	-0.504	—

Reading the table against the paper. In the paper the ad-stock coefficient falls when the control function enters at every level (0.023 to 0.010, 0.016 to 0.007, 0.018 to 0.012), and δ is positive and significant throughout. Here the moves are 0.0206 to 0.0021, 0.0246 to 0.0106, and 0.0274 to 0.0114 against a truth of 0.012 in the full model, with δ equal to 0.0203, 0.0101, and 0.0121. Columns 1 to 4 have no single true value: they fit one coefficient to an effect that the DGP makes vary with purchase history, so their β_0 is an average over states. The random-effect rows return $\sigma_1 = 0.184$ (DGP 0.2), $\sigma_2 = 0.016$ (0.01) and $\rho = -0.50$ (-0.3); the slope variance and the correlation have subsample bootstrap standard errors as large as the estimates, as in the paper, where neither is significant.

8.2 Elasticities: the headline number

The paper converts coefficients into elasticities to compare with the advertising literature. We define the same-day elasticity as the percent change in purchases when every household’s same-day exposures rise by one percent, $E[\partial P_{it}/\partial AS_{it} \cdot A_{it}]/E[P_{it}]$, with the derivative from `cf_ape()`, and the 30-day elasticity as the same-day value times $\sum_{l=0}^{29} \lambda_A^l$, the cumulative carryover of one day’s exposures.

```
f5 <- fit_column(pan, modifyList(specs$c5, list(random = FALSE)))
f6 <- fit_column(pan, modifyList(specs$c6, list(random = FALSE)))
el <- rbind(naive = elasticities(f5$fit, f5$data), corrected = elasticities(f6$fit, f6$data))
true_ape <- mean(dnorm(pan$index) * pan$beta_it)
el
#>               ape      same_day      long_run
#> naive      0.0008808777 0.017019699 0.05673105
#> corrected 0.0003435823 0.006868295 0.02289380
```

Table 10: Average partial effects and elasticities, naive against control function. The paper reports 0.072 and 0.222 without correction and 0.045 and 0.143 with it.

Model	Average partial effect of the ad stock	Same-day elasticity	30-day elasticity
Naive (column 5, fixed effects)	0.00088	0.017	0.057
Control function (column 6, fixed effects)	0.00034	0.007	0.023
Truth (DGP)	0.00029	—	—

The naive same-day elasticity overstates the corrected one by 148 percent (55 percent in the paper). The corrected average partial effect, 0.00034, is within sampling error of the DGP’s 0.00029.

9 Ad responsiveness and purchase history

Figure 4 is the paper’s substantive result: who responds to TV ads. It plots β_{it} implied by the column-6 estimates against frequency at fixed recency and against recency at fixed frequency. The bands come from the bootstrap draws.

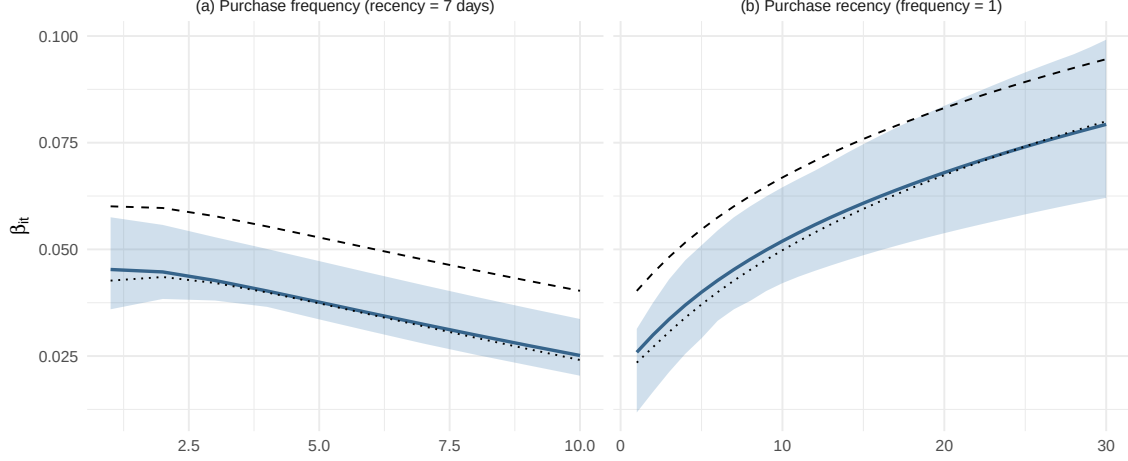


Figure 4: Ad responsiveness by purchase frequency and recency (the layout of Figure 4). Solid: control function (column 6) with 95 percent bootstrap bands; dashed: naive (column 5); dotted: the DGP.

Table 11: Baseline propensity and ad responsiveness by purchase state, column 6 estimates. The paper reports same-day elasticities of 0.040, 0.056, and 0.046 after one, two, and five purchases, and 0.040, 0.027, and 0.078 one day, one week, and four weeks after the last purchase.

State	Baseline daily purchase probability	Ad responsiveness β_{it}	Same-day elasticity
One prior purchase, 7 days ago	5.82%	0.0453	0.045
Two prior purchases, 7 days ago	7.74%	0.0447	0.041
Five prior purchases, 7 days ago	11.16%	0.0376	0.032
One prior purchase, 1 day ago	9.52%	0.0259	0.023
One prior purchase, 14 days ago	4.24%	0.0592	0.062
One prior purchase, 28 days ago	2.83%	0.0773	0.087

What the curves say. The DGP puts an inverted U in frequency ($\beta_{2,1} = 0.02$, $\beta_{2,2} = -0.01$) and a U in recency ($\beta_{2,4} = 0.005$), the paper’s two shapes. The control-function curve tracks the DGP within the band; the naive curve sits above it where the bias concentrates. The table is the paper’s targeting argument in numbers: lapsed households have a low baseline propensity and a high responsiveness, so they are worth an ad but only if it comes before the recency trap closes.

9.1 Table 5: observed household heterogeneity

Table 5 of the paper reports how the traits shift baseline propensity (α_1) and ad responsiveness (β_1). Heavy TV viewers order less and respond less; sports viewers order more and respond more. The DGP puts the same signs in with the same order of magnitude.

Table 12: Parameter estimates for observed household heterogeneity (the layout of Table 5), column 6. Standard errors from 50 bootstrap draws. ***p < 0.01; **p < 0.05; *p < 0.1.

Trait	Baseline purchase propensity (α_1)			Ad responsiveness (β_1)		
	True	Estimate	Standard error	True	Estimate	Standard error
Average TV viewing	-0.050	-0.046***	0.004	-0.004	-0.004**	0.002
Sports viewing	0.030	0.033***	0.003	0.004	0.002	0.001

The paper’s traits are nine viewing shares calibrated on the month before the study; the DGP has two. Their

coefficients are recovered with the signs and sizes put in, the responsiveness shifts less precisely because they multiply a stock whose effect is small.

10 The robustness checks the paper runs

10.1 The two decay rates (footnote 17)

The paper picks λ_A and λ_{CF} separately by out-of-sample fit. It estimates on the first 120 days, scores the log-likelihood on the remaining 13, and searches a grid. The two decays need not agree: the ad effect decays at the rate at which exposures are forgotten, the control function at the rate at which the confounder persists. The DGP sets 0.7 and 0.95. We run the grid on a subsample of 5,000 households.

```
grid_res <- file_cached("decay_grid", decay_grid(pan_sub, grid = c(0, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95), train_days = n_days - 30,
spec = col_spec(interactions = FALSE, traits = TRUE, trait_slopes = FALSE)))
grid_res[which.max(grid_res$loglik), ]
#>      lambda_A lambda_CF loglik
#> 42      0.95      0.9 -16964.02
```

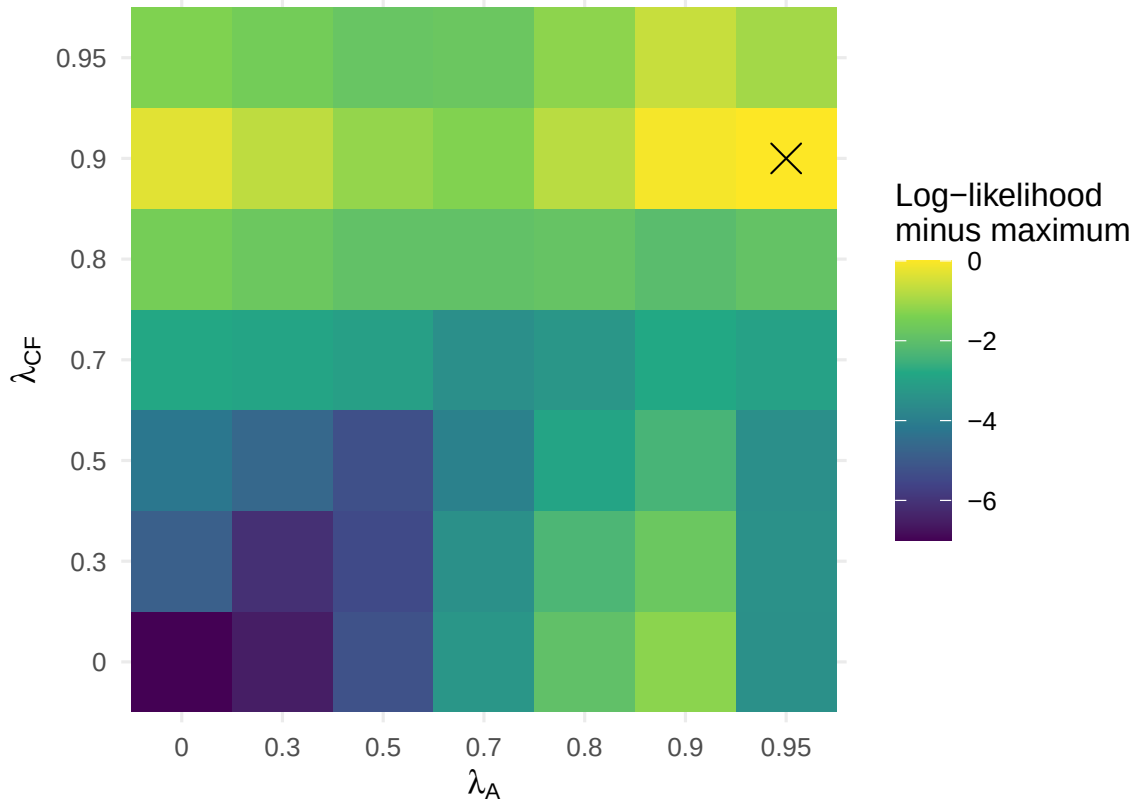


Figure 5: Out-of-sample log-likelihood over the grid of decay rates (the search of footnote 17); the cross marks the maximum, the DGP uses 0.7 and 0.95.

The maximum is at $\lambda_A = 0.95$, $\lambda_{CF} = 0.9$, against the DGP's 0.7 and 0.95, and the whole grid spans 7.0 log-likelihood points on 150,000 held-out observations. The criterion is flat near the top: a slower ad stock also proxies the slow-moving confounder when the specification omits the moderation terms, so the out-of-sample fit trades the two decays against each other. The paper's search faces the same weak identification, which is why it fixes the pair before comparing specifications rather than re-estimating it in each column.

10.2 Collinearity of the two stocks (footnote 18)

A stock and its control are built from the same daily series, so the paper reports variance inflation factors. Theirs are 2.42 and 2.27.

```
f4 <- fit_column(pan, specs$c4)
v <- car::vif(f4$fit)
c(cor_AS_CFS = cor(f4$data$A_stock, f4$data$CF_stock), vif_AS = unname(v["A_stock"]), vif_CFS = unname(v["CF_stock"]))
#> cor_AS_CFS      vif_AS      vif_CFS
#> 0.6232749  1.8345283  1.7017926
```

The two stocks correlate at 0.62 (0.72 in the paper) and the variance inflation factors are 1.83 and 1.70, below the paper's and far from the threshold of 10 that footnote 18 cites.

10.3 Placebo: another brand's ads (footnote 19, Online Appendix E.4)

If viewing is the confounder, any brand's exposures will predict purchases in a naive model. The paper replaces the focal brand's ads with a car maker's and finds no effect after correction. The simulated second brand reaches households through the same viewing process and has no effect on purchases.

```
pan_pl <- pan; pan_pl$A <- pan_pl$A_pl; pan_pl$IV <- pan_pl$IV_pl
pl_naive <- fit_column(pan_pl, specs$c3); pl_cf <- fit_column(pan_pl, specs$c4)
pl <- rbind(naive = summary(pl_naive$fit)$coefficients["A_stock", 1:2],
            control_function = summary(pl_cf$fit)$coefficients["A_stock", 1:2])
pl
#>               Estimate Std. Error
#> naive           0.02086972 0.001864229
#> control_function 0.00606142 0.002426620
```

The naive coefficient on the placebo stock is 0.0209 (standard error 0.0019); with the control function it is 0.0061 (standard error 0.0024). Activity bias alone produces an “effect” of ads that do not work, and the control function removes 71 percent of it. What remains is the approximation in the method: the confounder enters exposure through two nonlinear decisions, and a linear function of the first-stage residual captures most but not all of it.

10.4 Functional form of the history terms (Online Appendix E.1)

The paper compares four ways to let propensity and responsiveness bend with frequency and recency. Log plus squared log wins on AIC and BIC in their data; it is also the DGP's form here.

```
aug6 <- f6$data
base <- "purchase ~ A_stock + CF_stock + post + promo1m + comp + z_view + z_sports + A_stock:post"
forms <- list(
  linear = paste(base, "+ freq + rec + A_stock:freq + A_stock:rec"),
  quadratic = paste(base, "+ freq + I(freq^2) + rec + I(rec^2) + A_stock:freq + A_stock:I(freq^2)",
                    "+ A_stock:rec + A_stock:I(rec^2)"),
  log = paste(base, "+ lf + lr + A_stock:lf + A_stock:lr"),
  log_squared = paste(base, "+ lf + lf2 + lr + lr2 + A_stock:lf + A_stock:lf2 + A_stock:lr + A_stock:lr2")
)
ff <- file_cached("functional_forms", t(sapply(forms, function(f) {
  m <- glm(as.formula(f), data = aug6, family = binomial("probit"))
  c(loglik = as.numeric(logLik(m)), AIC = AIC(m), BIC = BIC(m), beta0 = unname(coef(m)["A_stock"])))
})))
cbind(round(ff[, 1:3], 1), beta0 = round(ff[, 4], 4))
#>           loglik      AIC      BIC  beta0
#> linear    -266151.2 532328.5 532499.3 0.0117
#> quadratic -265129.3 530292.6 530516.0 0.0123
#> log       -264398.8 528823.7 528994.5 0.0118
#> log_squared -264196.7 528427.4 528650.8 0.0125
```

Log plus squared log has the lowest AIC and BIC here as well, by 344 BIC points over the log specification; the ad-stock coefficient is the same to three decimals across the four forms.

11 Extensions

11.1 The instrument is a recentered instrument

Realized minus expected treatment under a known assignment process is the recentering of Borusyak and Hull (2023). Their point is that a shock-driven instrument is valid only after subtracting its expectation under the assignment process, because the expectation carries the nonrandom exposure. The paper’s instrument is that construction with the network’s ad-timing density as the process. The consequence is that the density must be right: Table 13 compares the paper’s instrument, a version that assumes uniform timing (expected exposure equals the viewed share), and the scripts’ oracle.

Table 13: Three instruments on the show-level panel: the paper’s, one that assumes uniform within-show timing, and the scripts’ oracle. Full model (column 6 without random effects). True $\beta_0 = 0.012$.

Instrument	Mean (t)	corr with A	corr with expected exposure	corr with u_{endo}	First-stage F	β_0	δ
Paper: empirical density	-0.0002 (-1.5)	0.292	-0.0004	-0.0002	353,612	0.0122	0.0102
Uniform timing (viewed share)	0.0142 (117.9)	0.340	0.0544	0.0232	498,117	0.0124	0.0105
Oracle: targeting probabilities	0.0001 (0.3)	0.611	0.4930	0.0002	2,280,935	0.0113	0.0160

The uniform version fails the falsification checks: its mean is not zero (t statistic of 0.0142 (117.9)) and it correlates with expected exposure at 0.0544 and with the unobservable at 0.0232, because the viewed share is not the exposure probability when the density is trimodal. Its second stage lands at 0.0124 against 0.0122 for the paper’s instrument, so at this size the leaked correlation is small enough for the control function to absorb; the check fails before the estimate does, which is the point of having a check. The oracle is valid but uses the targeting probabilities that no analyst observes.

11.2 What the diagnostics catch when the assumptions fail

Each identifying assumption has a diagnostic, and the Monte Carlo asks whether the diagnostic fires before the estimate is wrong. Strategic viewership: partial viewers with a positive demand shock skip the focal ad with probability π . Nonrandom ordering: the network puts the focal ad in the first pod with probability π . Small panels (2,000 households, 90 days), 20 replications each.

```
mc_grid <- expand_grid(type = c("skip", "order"), pi = c(0, 0.25, 0.5), stringsAsFactors = FALSE)
mc_grid <- mc_grid[!(mc_grid$pi == 0 & mc_grid$type == "order"), ]
mc <- file_cached(paste0("violation_mc_", n_sims), bind_rows(lapply(seq_len(nrow(mc_grid)), function(i) {
  violation_mc(n_sims = n_sims, n_hh = 2000, n_days = 90, type = if (mc_grid$pi[i] == 0) "none" else mc_grid$type[i],
    pi = mc_grid$pi[i], seed = i, workers = n_threads)
})))
mc_sum <- mc %>% mutate(type = ifelse(pi == 0, "none", type)) %>% group_by(type, pi) %>%
  summarise(n = n(), sd_cf = sd(beta_cf), beta_naive = mean(beta_naive), beta_cf = mean(beta_cf),
    ape_naive = mean(ape_naive), ape_cf = mean(ape_cf), ape_true = mean(ape_true), delta = mean(delta),
    iv_mean = mean(iv_mean), iv_t = mean(iv_t), corr_iv_exp = mean(corr_iv_exp),
    ks_reject = mean(ks_p_min < 0.05 / 3), .groups = "drop")
mc_sum
#> # A tibble: 5 x 14
#>   type    pi     n sd_cf beta_naive beta_cf ape_naive ape_cf ape_true delta
#>   <chr> <dbl> <int> <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
#> 1 none    0     20 0.0190    0.0310 0.0180 0.000740 4.66e-4 0.000427 0.00833
#> 2 order 0.25    20 0.0208    0.0298 0.0162 0.000691 4.01e-4 0.000436 0.00852
#> 3 order 0.5     20 0.0141    0.0354 0.0255 0.000802 5.98e-4 0.000420 0.00607
#> 4 skip   0.25    20 0.0229    0.0280 0.0162 0.000759 5.24e-4 0.000418 0.00707
#> 5 skip   0.5     20 0.0228    0.0380 0.0230 0.000764 4.59e-4 0.000417 0.00880
#> # 4 more variables: iv_mean <dbl>, iv_t <dbl>, corr_iv_exp <dbl>,
#> #   ks_reject <dbl>
```

Reading the table. Without a violation the corrected coefficient averages 0.0180 (truth 0.012, standard deviation across replications 0.0190) and the naive one 0.0310. Under strategic skipping, exposure falls short of its expectation for high-demand households, so the instrument’s mean turns negative with a t statistic of -108 at $\pi = 0.5$; the KS test cannot see it because the network did nothing. Under nonrandom ordering the

Table 14: Monte Carlo under violations of the identifying assumptions: estimates of the ad-stock coefficient (full model without traits; true value 0.012), average partial effects, and the diagnostics the paper proposes. Means over replications; the KS test rejects at 5 percent with a Bonferroni correction over the three networks.

Violation	π	Naive β_0	CF β_0	SD of CF β_0	APE naive / CF / true ($\times 10^4$)	IV mean (t)	corr(IV, expected)	KS rejects (share)
None	0.00	0.0310	0.0180	0.0190	7.40 / 4.66 / 4.27	-0.0000 (-0.0)	-0.000	0.00
Nonrandom ordering	0.25	0.0298	0.0162	0.0208	6.91 / 4.01 / 4.36	-0.0210 (-36.2)	0.030	1.00
Nonrandom ordering	0.50	0.0354	0.0255	0.0141	8.02 / 5.98 / 4.20	-0.0419 (-67.4)	0.056	1.00
Strategic viewership	0.25	0.0280	0.0162	0.0229	7.59 / 5.24 / 4.18	-0.0406 (-62.5)	0.006	0.15
Strategic viewership	0.50	0.0380	0.0230	0.0228	7.64 / 4.59 / 4.17	-0.0816 (-107.5)	0.011	0.05

KS test rejects in every replication and the mean-zero check also fails, because the density estimated from other brands no longer describes the focal ad. The corrected coefficient moves to 0.0230 under skipping and 0.0255 under reordering at $\pi = 0.5$. This is why the paper reports both checks and not one.

A direct effect of the instrument is the violation that no check detects. If the shifter affected purchases other than through exposure, the instrument would still be mean zero. Conley, Hansen, and Rossi (2012) bound the estimate under such a violation of size γ ; `iv_plausibly_exogenous()` does this for the linear-probability version.

```
pe <- iv_plausibly_exogenous(pan_sub, y = "purchase", d = "A", z = "IV", x = c(controls_base, controls_traits),
                             gamma_grid = c(-0.002, -0.001, 0, 0.001, 0.002), cluster = "hh")

pe
#>   gamma      estimate   std.error   conf.low   conf.high
#> 1 -0.002 0.0011179417 0.0007187466 -0.0002907758 0.0025266592
#> 2 -0.001 0.0001156173 0.0007186908 -0.0012929907 0.0015242254
#> 3  0.000 -0.0008867070 0.0007186518 -0.0022952386 0.0005218245
#> 4  0.001 -0.0018890314 0.0007186295 -0.0032975194 -0.0004805434
#> 5  0.002 -0.0028913557 0.0007186241 -0.0042998331 -0.0014828784
```

A direct effect of one exposure's shifter on the purchase probability of ± 0.002 , the size of the linear-probability estimate itself, widens the interval from $[-0.0023, 0.0005]$ to $[-0.0043, 0.0025]$. The mean-zero check cannot rule this violation out, only an argument about the mechanism can.

11.3 Estimators in probability units

The paper's control-function probit is one of three ways to use the instrument. Two-stage least squares on the linear-probability model instruments the ad stock with the instrument stock; the partially linear IV model of double machine learning does the same with flexible nuisance functions of the states and traits (`est_dml(model = "pliv")`, forests on a subsample); the control function gives an average partial effect through `cf_ape()`. All three are in probability units and can be compared with the DGP.

```
xs <- c(controls_base, controls_traits)
xs_sum <- paste(xs, collapse = " + ")
est_cache <- file_cached("estimators", {
  tsls <- iv_robust(as.formula(paste("purchase ~ A_stock +", xs_sum, "| IV_stock +", xs_sum)), data = pan,
                   clusters = hh, se_type = "stata")
  sub2 <- pan[pan$hh %in% sub_hh[seq_len(2000L)], ]
  rf <- lrn("regr.ranger", num.trees = 200, min.node.size = 50, num.threads = n_threads)
  dml <- est_dml(sub2, y = "purchase", d = "A_stock", x = xs, z = "IV_stock", model = "pliv",
                 learner_l = rf, learner_m = rf, learner_z = rf, folds = 2, seed = 1)
  list(tsls_est = unname(coef(tsls)["A_stock"]), tsls_se = unname(tsls$std.error["A_stock"]),
       dml_est = dml$estimate, dml_se = dml$std.error)
})
ape_cf <- cf_ape(f6$fit, f6$data, d = "A_stock")
se_ape <- b4$c6$se["A_stock"] * abs(as.numeric(ape_cf) / unname(t4$c6$coef["A_stock"]))
est_tab <- tibble(
  Estimator = c("2SLS, linear probability (household clusters)", "DML partially linear IV, forests (2,000 households)",
                "Control-function probit APE (column 6)", "Truth (DGP)"),
  Estimate = c(est_cache$tsls_est, est_cache$dml_est, as.numeric(ape_cf), true_ape),
  SE = c(est_cache$tsls_se, est_cache$dml_se, se_ape, NA)
)
est_tab
```

```
#> # A tibble: 4 x 3
#>   Estimator      Estimate      SE
#>   <chr>      <dbl>      <dbl>
#> 1 2SLS, linear probability (household clusters) -0.0000467 0.000239
#> 2 DML partially linear IV, forests (2,000 households) -0.000101 0.000785
#> 3 Control-function probit APE (column 6) 0.000344 0.000103
#> 4 Truth (DGP) 0.000290 NA
```

Table 15: Average effect of one unit of ad stock on the daily purchase probability, three estimators against the DGP. The control-function standard error scales the bootstrap standard error of the coefficient.

Estimator	Estimate	SE
2SLS, linear probability (household clusters)	-0.00005	0.00024
DML partially linear IV, forests (2,000 households)	-0.00010	0.00079
Control-function probit APE (column 6)	0.00034	0.00010
Truth (DGP)	0.00029	—

The linear estimators do not estimate the average partial effect here. Two-stage least squares lands at -0.00005, the marginal effect of a household before conversion (0.00006 on average), not the DGP’s 0.00029. With one coefficient on the ad stock and linear controls, 2SLS averages the state-specific effects with weights that put almost all mass on the 88 percent of household-days before conversion, where the purchase probability and with it the effect are tiny, and the linear controls cannot follow a probability that jumps by a factor of a hundred at conversion. DML-IV on 2,000 households replaces the linear controls with forests but is too imprecise here to separate the two. The probit control function averages $\Phi'(\cdot) \beta_{it}$ over the observed states and matches the DGP. The lesson for a rare-outcome panel with state dependence: the estimand of a linear IV is a variance-weighted average that need not be the average partial effect, and the nonlinear second stage is not a cosmetic choice.

11.4 Inference for the two-step estimator

Three standard errors for the same coefficients. The model-based one treats the control-function stock as data; the authors’ bootstrap keeps unique households; the cluster bootstrap keeps duplicates. Table 16 puts them side by side for the benchmark of Section 4 (authors’ DGP) and for column 4 of Table 9 (show-level panel).

Table 16: Model-based, unique-household bootstrap, and cluster bootstrap standard errors of the control-function probit.

Panel	Term	Model-based	Authors’ bootstrap	Cluster bootstrap
Authors’ DGP (full model)	Ad stock	0.0026	0.0024	0.0029
	Control function	0.0006	0.0004	0.0006
	Post conversion	0.0168	0.0127	0.0180
	First month post-conversion promotion	0.0040	0.0030	0.0045
Show-level panel (column 4)	Ad stock	0.0023	—	0.0024
	Control function	0.0011	—	0.0010
	Post conversion	0.0185	—	0.0182
	First month post-conversion promotion	0.0070	—	0.0081

The model-based standard error of the ad-stock coefficient is 0.88 of the cluster bootstrap’s on the authors’ DGP and 0.94 on the show-level panel; the generated regressor is the reason. The authors’ bootstrap is 0.73 of the cluster bootstrap on average, the 0.76 of a 63 percent subsample.

12 What the replication teaches about the workflow

- **The instrument is a design, not a variable.** It is built from the assignment process (the network's density) and the observed viewing window, and it is testable in three ways that a fitted propensity score is not. The mean-zero and orthogonality checks of Table 2 are the evidence to show; corr with u cannot be shown and is what the checks stand in for.
- **A show level is required to build it.** The authors' scripts simulate the household-day outcome of the design with an oracle expectation. Anyone applying the method needs second-by-second viewing and all brands' airing times, not exposures alone.
- **Control functions need stocks when regressors do.** The confounder persists at its own rate, so the control gets its own decay, chosen out of sample. `cf_residuals()` gives the residual; two lines accumulate it.
- **Bootstrap the whole pipeline, by cluster, with duplicates.** The model-based standard errors are too small for the generated regressor, and a bootstrap that keeps unique households is a subsample. The relabelling of duplicated households is what makes a stock accumulate correctly inside a cluster bootstrap.
- **The diagnostics are specific.** Strategic viewership shows in the instrument's mean; nonrandom ordering shows in the KS test; a direct effect shows in neither and needs a sensitivity bound.
- **Random coefficients are the expensive part.** Columns 5 and 6 with household random effects are the only fits in this note that take longer than a minute; the Laplace approximation of `glmer()` reproduces the structure of the paper's simulated maximum likelihood at a fraction of the cost, and the bootstrap of column 6 is the compute budget of the whole note.

13 References

- Borusyak, K. and Hull, P. (2023). Nonrandom exposure to exogenous shocks. *Econometrica*, 91(6), 2155-2185.
- Conley, T. G., Hansen, C. B., and Rossi, P. E. (2012). Plausibly exogenous. *Review of Economics and Statistics*, 94(1), 260-272.
- Hsieh, T.-Y., Du, R. Y., and Lu, S. (2026). Leveraging large-scale granular single-source data for TV advertising: an identification strategy. *Marketing Science*, articles in advance.
- Petrin, A. and Train, K. (2010). A control function approach to endogeneity in consumer choice models. *Journal of Marketing Research*, 47(1), 3-13.
- Sethuraman, R., Tellis, G. J., and Briesch, R. A. (2011). How well does advertising work? Generalizations from a meta-analysis of brand advertising elasticity. *Journal of Marketing Research*, 48(3), 457-471.
- Wooldridge, J. M. (2015). Control function methods in applied econometrics. *Journal of Human Resources*, 50(2), 420-445.